

Complementary low voltage transistor

Features

- Products are pre-selected in DC current gain

Application

- General purpose

Description

These epitaxial planar transistors are mounted in the SOT-32 plastic package. They are designed for audio amplifiers and drivers utilizing complementary or quasi-complementary circuits. The NPN types are the BD135 and BD139, and the complementary PNP types are the BD136 and BD140.

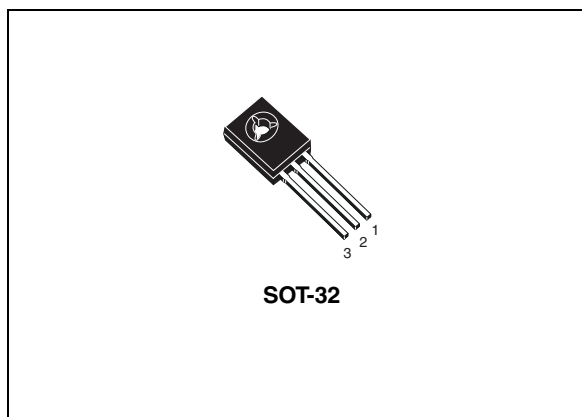


Figure 1. Internal schematic diagram

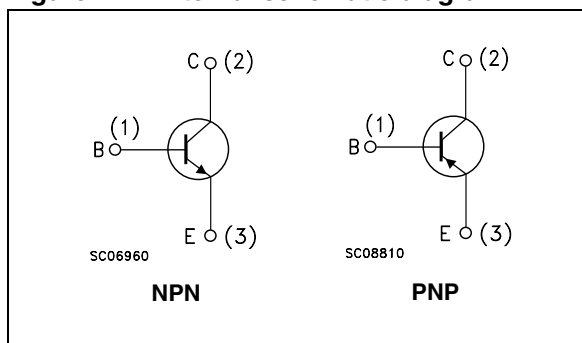


Table 1. Device summary

Order codes	Marking	Package	Packaging
BD135	BD135	SOT-32	Tube
BD135-16	BD135-16		
BD136	BD136		
BD136-16	BD136-16		
BD139	BD139		
BD139-10	BD139-10		
BD139-16	BD139-16		
BD140	BD140		
BD140-10	BD140-10		
BD140-16	BD140-16		

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1 Electrical ratings

Table 2. Absolute maximum ratings

Symbol	Parameter	Value				Unit
		NPN		PNP		
		BD135	BD139	BD136	BD140	
V _{CBO}	Collector-base voltage (I _E = 0)	45	80	-45	-80	V
V _{CEO}	Collector-emitter voltage (I _B = 0)	45	80	-45	-80	V
V _{EBO}	Emitter-base voltage (I _C = 0)	5		-5		V
I _C	Collector current	1.5		-1.5		A
I _{CM}	Collector peak current	3		-3		A
I _B	Base current	0.5		-0.5		A
P _{TOT}	Total dissipation at T _c ≤ 25 °C	12.5				W
P _{TOT}	Total dissipation at T _{amb} ≤ 25 °C	1.25				W
T _{stg}	Storage temperature	-65 to 150				°C
T _j	Max. operating junction temperature	150				°C

Table 3. Thermal data

Symbol	Parameter	Max value	Unit
$R_{thj-case}$	Thermal resistance junction-case	10	°C/W
$R_{thj-amb}$	Thermal resistance junction-ambient	100	°C/W

2 Electrical characteristics

($T_{case} = 25\text{ °C}$ unless otherwise specified)

Table 4. On/off states

Symbol	Parameter	Polarity	Test conditions	Value			Unit
				Min.	Typ.	Max.	
I_{CBO}	Collector cut-off current ($I_E=0$)	NPN	$V_{CB} = 30\text{ V}$ $V_{CB} = 30\text{ V}, T_C = 125\text{ °C}$			0.1 10	μA μA
		PNP	$V_{CB} = -30\text{ V}$ $V_{CB} = -30\text{ V}, T_C = 125\text{ °C}$			-0.1 -10	μA μA
I_{EBO}	Emitter cut-off current ($I_C=0$)	NPN	$V_{EB} = 5\text{ V}$			10	μA
		PNP	$V_{EB} = -5\text{ V}$			-10	μA
$V_{CEO(sus)}^{(1)}$	Collector-emitter sustaining voltage ($I_B=0$)	NPN	$I_C = 30\text{ mA}$ BD135 BD139	45 80			V V
		PNP	$I_C = -30\text{ mA}$ BD136 BD140	-45 -80			V V
$V_{CE(sat)}^{(1)}$	Collector-emitter saturation voltage	NPN	$I_C = 0.5\text{ A}, I_B = 0.05\text{ A}$			0.5	V
		PNP	$I_C = -0.5\text{ A}, I_B = -0.05\text{ A}$			-0.5	V
$V_{BE}^{(1)}$	Base-emitter voltage	NPN	$I_C = 0.5\text{ A}, V_{CE} = 2\text{ V}$			1	V
		PNP	$I_C = -0.5\text{ A}, V_{CE} = -2\text{ V}$			-1	V
$h_{FE}^{(1)}$	DC current gain	NPN	$I_C = 5\text{ mA}, V_{CE} = 2\text{ V}$ $I_C = 150\text{ mA}, V_{CE} = 2\text{ V}$ $I_C = 0.5\text{ A}, V_{CE} = 2\text{ V}$	25 40 25		250	
		PNP	$I_C = -5\text{ mA}, V_{CE} = -2\text{ V}$ $I_C = -150\text{ mA}, V_{CE} = -2\text{ V}$ $I_C = -0.5\text{ A}, V_{CE} = -2\text{ V}$	25 40 25		250	
$h_{FE}^{(1)}$	h_{FE} groups	NPN	$I_C = 150\text{ mA}, V_{CE} = 2\text{ V}$ BD139-10 BD135-16/BD139-16	63 100		160 250	
		PNP	$I_C = -150\text{ mA}, V_{CE} = -2\text{ V}$ BD140-10 BD136-16/BD140-16	63 100		160 250	

1. Pulsed: pulse duration = 300 μs , duty cycle 1.5%

2.1 Electrical characteristics (curves)

Figure 2. Safe operating area

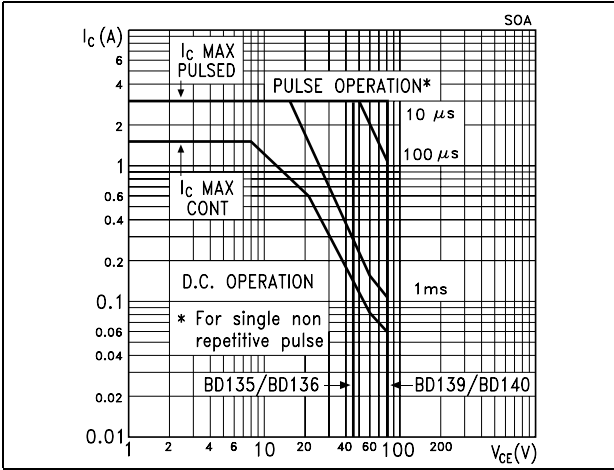
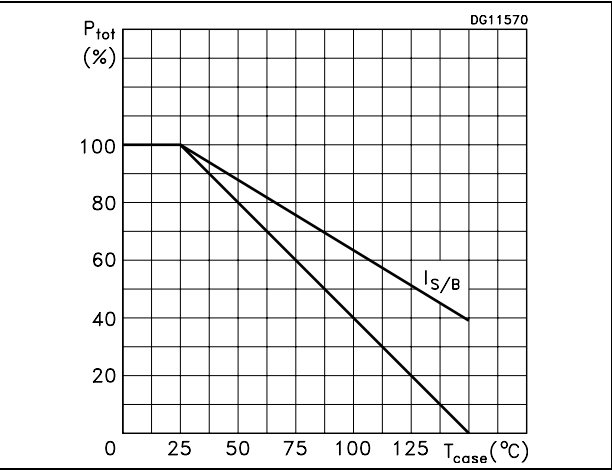


Figure 3. Derating

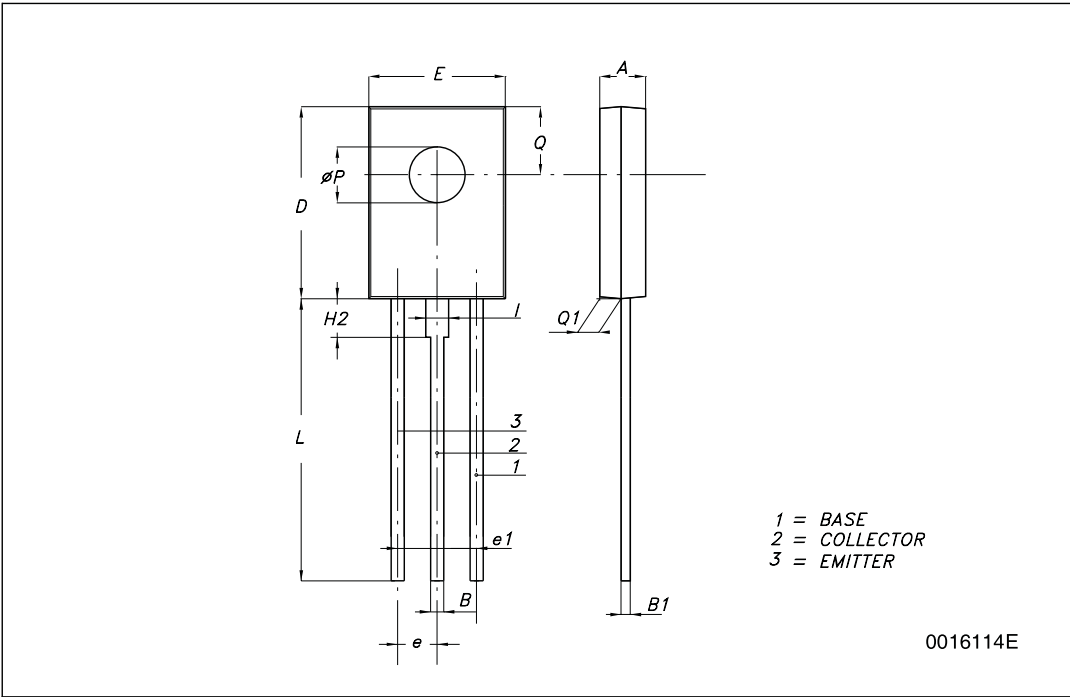


3 Package mechanical data

In order to meet environmental requirements, ST offers these devices in ECOPACK® packages. These packages have a lead-free second level interconnect. The category of second level interconnect is marked on the package and on the inner box label, in compliance with JEDEC Standard JESD97. The maximum ratings related to soldering conditions are also marked on the inner box label. ECOPACK is an ST trademark. ECOPACK specifications are available at: www.st.com

SOT-32 (TO-126) MECHANICAL DATA

DIM.	mm.		
	MIN.	TYP	MAX.
A	2.4		2.9
B	0.64		0.88
B1	0.39		0.63
D	10.5		11.05
E	7.4		7.8
e	2.04	2.29	2.54
e1	4.07	4.58	5.08
L	15.3		16
P	2.9		3.2
Q		3.8	
Q1	1		1.52
H2		2.15	
I		1.27	



4 Revision history

Table 5. Document revision history

Date	Revision	Changes
16-Sep-2001	4	
22-May-2008	5	Mechanical data has been updated.

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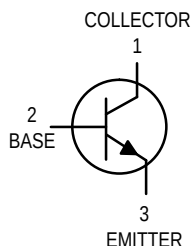
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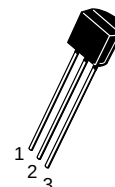
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Amplifier Transistors

NPN Silicon



BC546, B
BC547, A, B, C
BC548, A, B, C



CASE 29-04, STYLE 17
TO-92 (TO-226AA)

MAXIMUM RATINGS

Rating	Symbol	BC 546	BC 547	BC 548	Unit
Collector–Emitter Voltage	V_{CEO}	65	45	30	Vdc
Collector–Base Voltage	V_{CBO}	80	50	30	Vdc
Emitter–Base Voltage	V_{EBO}	6.0			Vdc
Collector Current — Continuous	I_C	100			mA dc
Total Device Dissipation @ $T_A = 25^\circ\text{C}$ Derate above 25°C	P_D	625 5.0			mW mW/°C
Total Device Dissipation @ $T_C = 25^\circ\text{C}$ Derate above 25°C	P_D	1.5 12			Watt mW/°C
Operating and Storage Junction Temperature Range	T_J, T_{stg}	–55 to +150			°C

THERMAL CHARACTERISTICS

Characteristic	Symbol	Max	Unit
Thermal Resistance, Junction to Ambient	$R_{\theta JA}$	200	°C/W
Thermal Resistance, Junction to Case	$R_{\theta JC}$	83.3	°C/W

ELECTRICAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$ unless otherwise noted)

Characteristic	Symbol	Min	Typ	Max	Unit
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OFF CHARACTERISTICS

Collector–Emitter Breakdown Voltage ($I_C = 1.0\text{ mA}$, $I_B = 0$)	BC546 BC547 BC548	$V_{(BR)CEO}$	65 45 30	— — —	— — —	V
Collector–Base Breakdown Voltage ($I_C = 100\text{ }\mu\text{A dc}$)	BC546 BC547 BC548	$V_{(BR)CBO}$	80 50 30	— — —	— — —	V
Emitter–Base Breakdown Voltage ($I_E = 10\text{ }\mu\text{A}$, $I_C = 0$)	BC546 BC547 BC548	$V_{(BR)EBO}$	6.0 6.0 6.0	— — —	— — —	V
Collector Cutoff Current ($V_{CE} = 70\text{ V}$, $V_{BE} = 0$) ($V_{CE} = 50\text{ V}$, $V_{BE} = 0$) ($V_{CE} = 35\text{ V}$, $V_{BE} = 0$) ($V_{CE} = 30\text{ V}$, $T_A = 125^\circ\text{C}$)	BC546 BC547 BC548 BC546/547/548	I_{CES}	— — — —	0.2 0.2 0.2 —	15 15 15 4.0	nA μA



BC546, B BC547, A, B, C BC548, A, B, C**ELECTRICAL CHARACTERISTICS** ($T_A = 25^\circ\text{C}$ unless otherwise noted) (Continued)

Characteristic	Symbol	Min	Typ	Max	Unit
ON CHARACTERISTICS					
DC Current Gain ($I_C = 10\ \mu\text{A}$, $V_{CE} = 5.0\ \text{V}$)	h_{FE}	—	90	—	—
BC547A/548A		—	150	—	—
BC546B/547B/548B BC548C		—	270	—	—
($I_C = 2.0\ \text{mA}$, $V_{CE} = 5.0\ \text{V}$)	h_{FE}	110	—	450	—
BC546		110	—	800	—
BC547		110	—	800	—
BC548		110	180	220	—
BC547A/548A		200	290	450	—
BC546B/547B/548B BC547C/BC548C		420	520	800	—
($I_C = 100\ \text{mA}$, $V_{CE} = 5.0\ \text{V}$)	h_{FE}	—	120	—	—
BC547A/548A		—	180	—	—
BC546B/547B/548B BC548C		—	300	—	—
Collector–Emitter Saturation Voltage ($I_C = 10\ \text{mA}$, $I_B = 0.5\ \text{mA}$)	$V_{CE(sat)}$	—	0.09	0.25	V
($I_C = 100\ \text{mA}$, $I_B = 5.0\ \text{mA}$)		—	0.2	0.6	V
($I_C = 10\ \text{mA}$, $I_B = \text{See Note 1}$)		—	0.3	0.6	V
Base–Emitter Saturation Voltage ($I_C = 10\ \text{mA}$, $I_B = 0.5\ \text{mA}$)	$V_{BE(sat)}$	—	0.7	—	V
Base–Emitter On Voltage ($I_C = 2.0\ \text{mA}$, $V_{CE} = 5.0\ \text{V}$)	$V_{BE(on)}$	0.55	—	0.7	V
($I_C = 10\ \text{mA}$, $V_{CE} = 5.0\ \text{V}$)		—	—	0.77	V

SMALL–SIGNAL CHARACTERISTICS

Current–Gain — Bandwidth Product ($I_C = 10\ \text{mA}$, $V_{CE} = 5.0\ \text{V}$, $f = 100\ \text{MHz}$)	f_T	150	300	—	MHz
BC546		150	300	—	MHz
BC547 BC548		150	300	—	MHz
Output Capacitance ($V_{CB} = 10\ \text{V}$, $I_C = 0$, $f = 1.0\ \text{MHz}$)	C_{obo}	—	1.7	4.5	pF
Input Capacitance ($V_{EB} = 0.5\ \text{V}$, $I_C = 0$, $f = 1.0\ \text{MHz}$)	C_{ibo}	—	10	—	pF
Small–Signal Current Gain ($I_C = 2.0\ \text{mA}$, $V_{CE} = 5.0\ \text{V}$, $f = 1.0\ \text{kHz}$)	h_{fe}	125	—	500	—
BC546		125	—	900	—
BC547/548		125	220	260	—
BC547A/548A		240	330	500	—
BC546B/547B/548B BC547C/548C		450	600	900	—
Noise Figure ($I_C = 0.2\ \text{mA}$, $V_{CE} = 5.0\ \text{V}$, $R_S = 2\ \text{k}\Omega$, $f = 1.0\ \text{kHz}$, $\Delta f = 200\ \text{Hz}$)	NF	—	2.0	10	dB
BC546		—	2.0	10	dB
BC547 BC548		—	2.0	10	dB

Note 1: I_B is value for which $I_C = 11\ \text{mA}$ at $V_{CE} = 1.0\ \text{V}$.

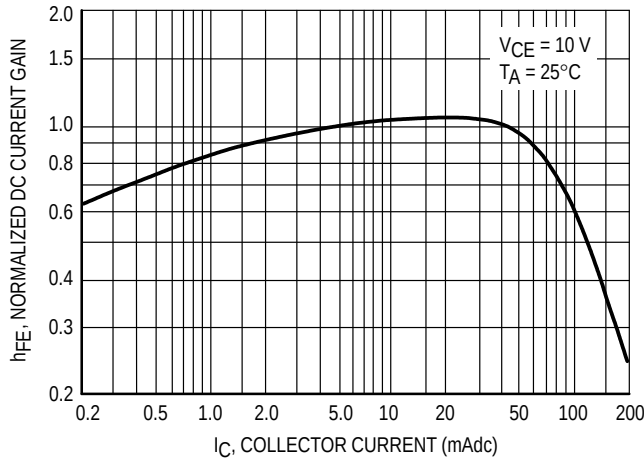


Figure 1. Normalized DC Current Gain

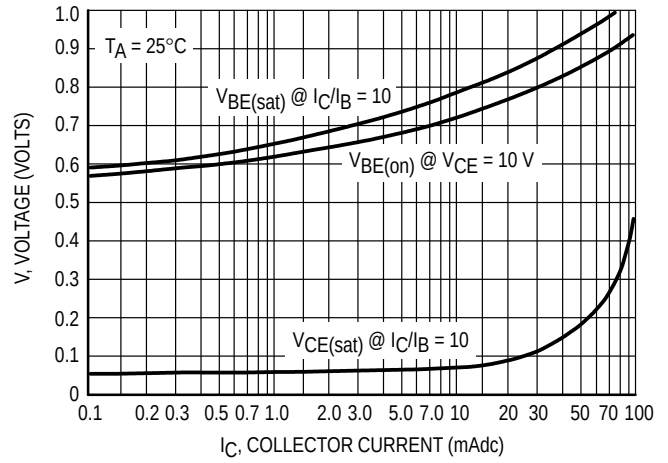


Figure 2. "Saturation" and "On" Voltages

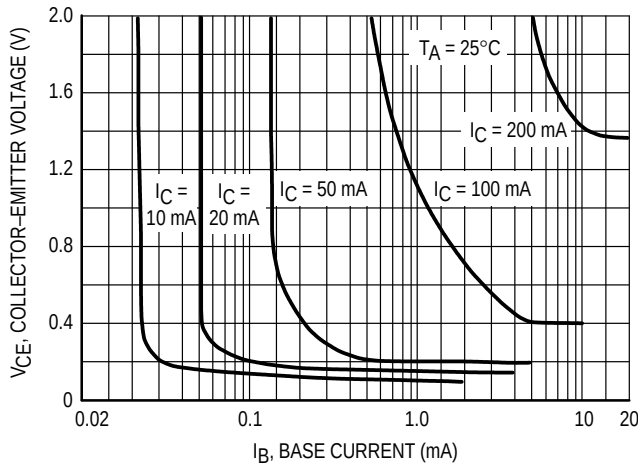


Figure 3. Collector Saturation Region

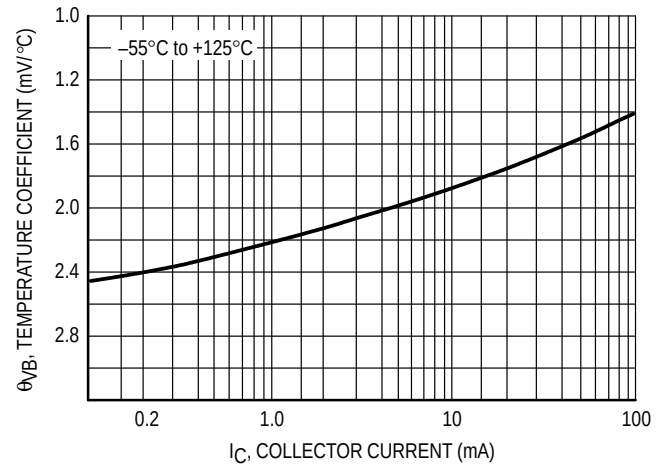


Figure 4. Base-Emitter Temperature Coefficient

BC547/BC548

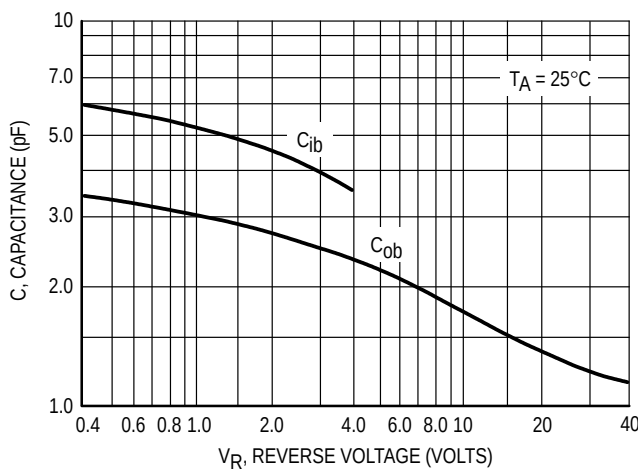


Figure 5. Capacitances

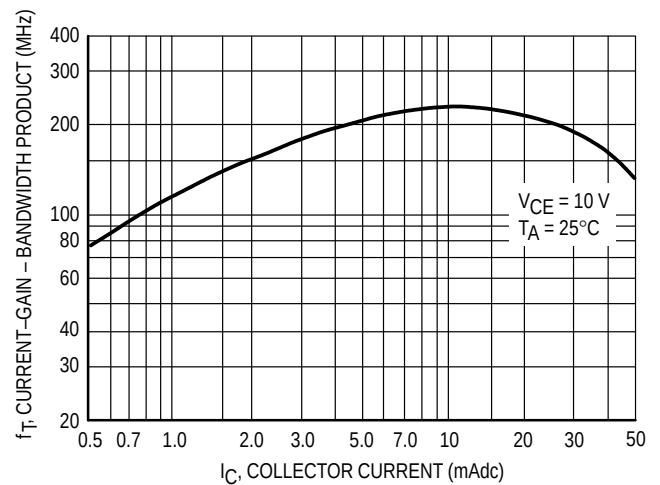


Figure 6. Current-Gain - Bandwidth Product

BC547/BC548

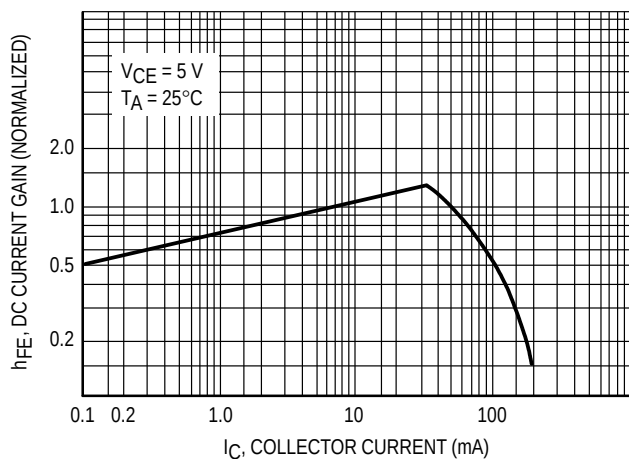


Figure 7. DC Current Gain

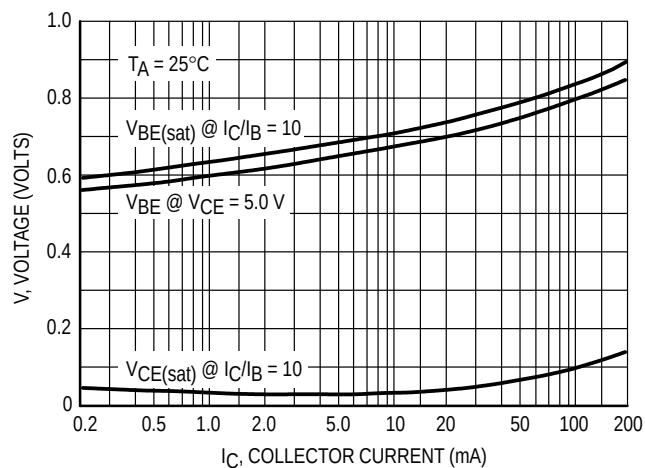


Figure 8. "On" Voltage

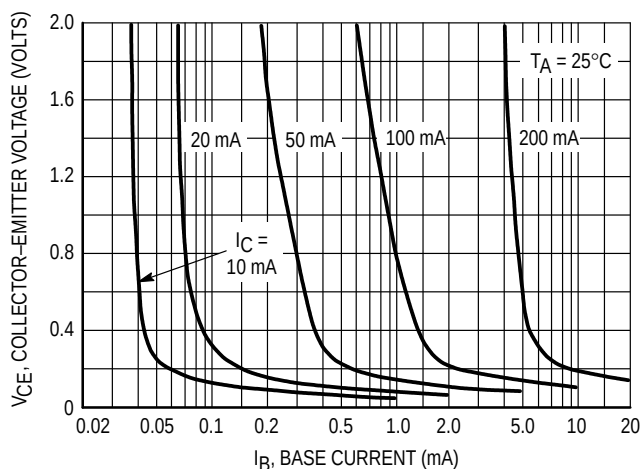


Figure 9. Collector Saturation Region

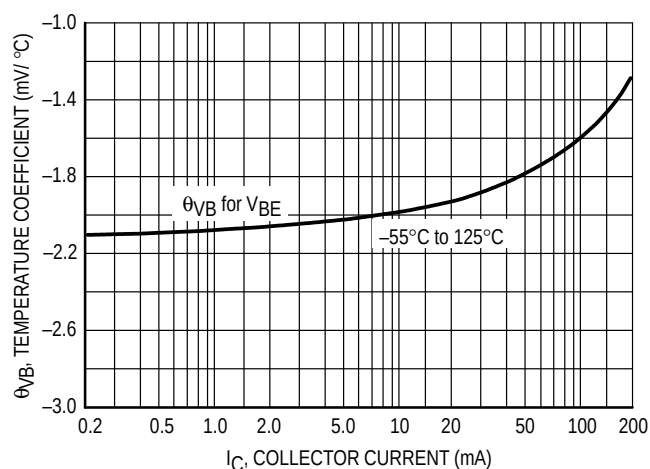


Figure 10. Base-Emitter Temperature Coefficient

BC546

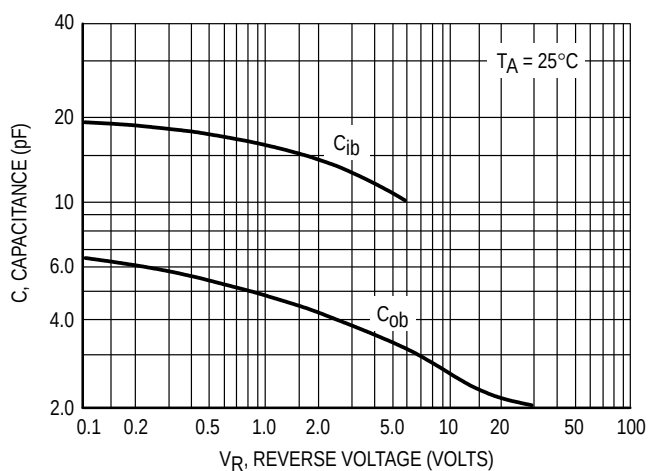


Figure 11. Capacitance

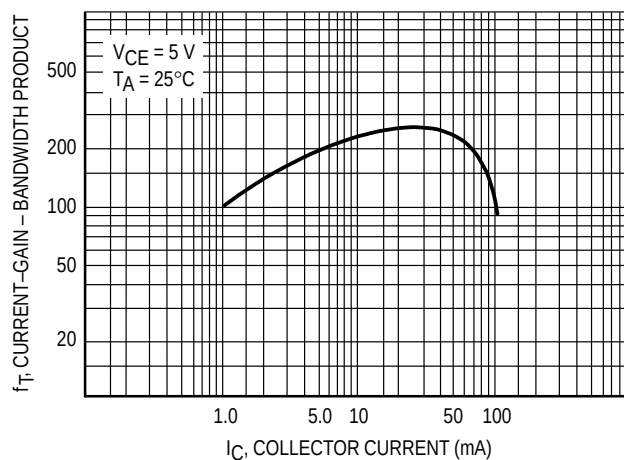
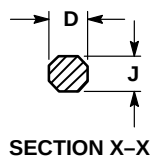
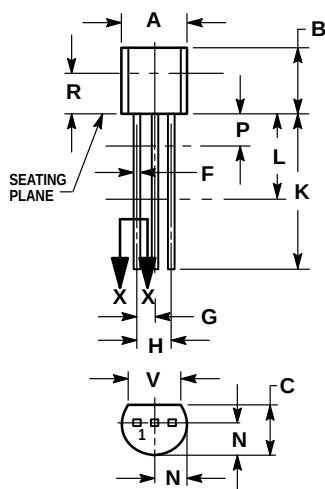


Figure 12. Current-Gain - Bandwidth Product

PACKAGE DIMENSIONS



**CASE 029-04
(TO-226AA)
ISSUE AD**

NOTES:

1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M, 1982.
2. CONTROLLING DIMENSION: INCH.
3. CONTOUR OF PACKAGE BEYOND DIMENSION R IS UNCONTROLLED.
4. DIMENSION F APPLIES BETWEEN P AND L. DIMENSION D AND J APPLY BETWEEN L AND K MINIMUM. LEAD DIMENSION IS UNCONTROLLED IN P AND BEYOND DIMENSION K MINIMUM.

DIM	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
A	0.175	0.205	4.45	5.20
B	0.170	0.210	4.32	5.33
C	0.125	0.165	3.18	4.19
D	0.016	0.022	0.41	0.55
F	0.016	0.019	0.41	0.48
G	0.045	0.055	1.15	1.39
H	0.095	0.105	2.42	2.66
J	0.015	0.020	0.39	0.50
K	0.500	—	12.70	—
L	0.250	—	6.35	—
N	0.080	0.105	2.04	2.66
P	—	0.100	—	2.54
R	0.115	—	2.93	—
V	0.135	—	3.43	—

STYLE 17:

1. COLLECTOR
2. BASE
3. EMITTER

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BC546/D

